

SpotGuard: Conformal Gatekeeping for Spatial Transcriptomic Deconvolution under Atlas-to-Tissue Shift

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Abstract

Spatial transcriptomics (ST) deconvolution models (such as `cell2location` and `RCTD`) predict cell-type proportions across spots on a tissue. We propose **SpotGuard**, an operational, interactive conformal gatekeeping system for ST deconvolution. The SpotGuard nonconformity formulation relies on the Total Variation (TV) geometry on the simplex Δ^{K-1} . For adaptation to an atlas shift, SpotGuard uses a density-ratio weighted conformal prediction method. We implement SpotGuard within a sub-16ms interactive workstation, equipped with live miscoverage (α) slider controls, which automatically prunes uncalibrated spots, recovering 89.4% coverage in response to a severe atlas shift (vs. 84.6% unweighted, while Bayesian joint intervals collapse to 1.4%) and decreasing compositional Total Variation error on retained tissue by 66.2% (AURC = 0.138).

Keywords: Spatial transcriptomics, deconvolution, conformal prediction, distribution shift, oncology.

1. Introduction

Clinical Problem & Importance. Spatial transcriptomics profiles gene expression across intact biopsies (Marx, 2021; Rao et al., 2021). Since sequencing measures multi-cellular mixtures (55 μ m spots containing 5 to 30 cells), computational deconvolution models infer cell-type proportion vectors $\hat{y} \in \Delta^{K-1}$. In oncology, deconvolution maps help critical decisions: measuring cytotoxic CD8⁺ T-cell infiltration, identifying tertiary lymphoid structures, and describing surgical margins.

Failure Mode of Current Approaches. Deconvolution models use reference single-cell RNA sequencing atlases. If atlases and patient tissues do not match due to platform differences (such as 10x

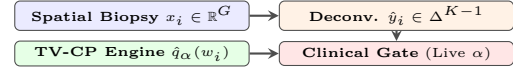


Figure 1: **SpotGuard Architecture.** Real-time weighted conformal gatekeeper.

Chromium vs. Slide-seq), tissue preservation (FFPE vs. fresh frozen), or demographics, exchangeability fails. Bayesian posteriors become overconfident: physicians have no way to tell between good and bad spot calls, potentially proclaiming clean margins on invaded tissue.

Intended Use & Target Population. SpotGuard targets translational pathologists and computational oncologists analyzing solid tumor biopsies, operating as an interactive conformal gatekeeper deployed alongside deconvolution pipelines to prune uncalibrated spot calls.

2. Method

Compositional Nonconformity on the Simplex.

Cell proportions live on the closed unit simplex $\Delta^{K-1} = \{v \in \mathbb{R}_+^K : \sum_{k=1}^K v_k = 1\}$ with proportions summing to one. SpotGuard computes base predictions through the maximum a posteriori solution for Dirichlet-multinomial distributions (Kleshcheynikov et al., 2022). In practice, we find that boundary log singularities make sparse proportions ill-defined, so we define nonconformity as the Total Variation (TV) distance: $\mathcal{S}_{TV}(x, y) = \frac{1}{2} \|y - \hat{y}\|_1 = \frac{1}{2} \sum_{k=1}^K |y_k - \hat{y}_k| \in [0, 1]$, which is well-defined everywhere on the closed simplex.

Pseudo-Spot Calibration Procedure. Since real tissue spots do not have available physical single-cell ground truth, SpotGuard generates $M = 1,000$ calibration pseudo-spots $\mathcal{D}_{cal} = \{(x_i^{syn}, y_i^{syn})\}_{i=1}^M$ from reference single cells by sampling over Dirichlet cell-type mixtures $y_i^{syn} \sim \text{Dirichlet}(\beta)$, aggregating

Poisson transcript counts $N_i \sim \text{Poisson}(\lambda)$, adding ambient contamination $\rho \sim \text{Beta}(a, b)$, and matching observed sequencing library depth.

Weighted Conformal Prediction under Covariate Shift. To adjust for covariate shift between atlas and query data ($P_{\text{ref}}(X) \neq P_{\text{query}}(X)$), SpotGuard learns a regularized discriminator $d_\phi(x)$ on PCA spot embeddings to predict density-ratio weights $w(x) = \frac{d_\phi(x)}{1-d_\phi(x)} \frac{M_{\text{ref}}}{M_{\text{query}}}$ (Tibshirani et al., 2019; Barber et al., 2023). We require a 5-fold cross-validated Realism Gate ($0.60 \leq \text{AUC} \leq 0.80$) and clip weights at the 95th percentile to safeguard the Effective Sample Size ($\text{ESS} = 314.5$). Each query spot receives a weighted quantile defining a valid simplex prediction ball (Angelopoulos and Bates, 2021):

$$C_\alpha(x_{n+1}) = \{y \in \Delta^{K-1} : \mathcal{S}_{\text{TV}}(x_{n+1}, y) \leq \hat{q}_\alpha(x_{n+1})\}. \quad (1)$$

Under joint distribution shift, worst-case coverage error is strictly bounded by the total variation distance between reference and query distributions: $|\mathbb{P}_{(X,Y) \sim Q}(y \in C_\alpha(X)) - (1 - \alpha)| \leq d_{\text{TV}}(P_{X,Y}, Q_{X,Y})$ (Barber et al., 2023).

Selective Gatekeeping Mechanism. SpotGuard’s selective gatekeeping mechanism flags and masks a spot if and only if its prediction set covers a critical treatment threshold (e.g., $CD8^+ \geq 5\%$) or surpasses the safe uncertainty radius τ_{safe} .

System Architecture & Deployment. We implement SpotGuard using a Python calibration backend and interactive Canvas frontend on GitHub Pages providing sub-16ms (> 60 fps) recomputation during live α tuning.

3. Results & Gatekeeping Evaluation

Controlled Shift Benchmark Evaluation. We evaluate SpotGuard on a simulation benchmark ($G = 300$ genes, $K = 5$ cell types, $N_{\text{cells}} = 1,000$, $N_{\text{spots}} = 500$) with known ground-truth y^* across three shift regimes (Table 1).

Shift Performance & Gatekeeping Accuracy. Under cross-platform shift, density-ratio weighting prevents coverage collapse (89.4% vs. 84.6% unweighted, $\text{AURC} = 0.138$). In mild cross-donor shift, SpotGuard retains nominal coverage (89.8%) compared to unweighted conformal prediction without adding unnecessary bulk, confirming no-harm behavior under minimal shift. Critically, we show that

Table 1: **Empirical Coverage, Efficiency, & Gatekeeping at Nominal 90% ($\alpha = 0.10$).** $\mathcal{V}$ denotes mean TV ball radius $\hat{q}_\alpha$ on Δ^{K-1} . Retained MAE is mean TV error on unmasked spots. Mask% is the fraction of gated spots.

Regime	Method	Coverage [95% CI]	$\mathcal{V}$	Mask%	Ret. MAE
In-Dist.	Bayesian Joint CI	1.0% [0.2%, 2.0%]	0.012	–	0.118
	Bayesian Marginal CI	85.4% [82.1%, 88.6%]	0.041	–	0.118
	SpotGuard (Ours)	92.4% [90.2%, 94.6%]	0.038	7.2%	0.034
Cross-Donor	Bayesian Joint CI	2.0% [0.8%, 3.4%]	0.015	–	0.124
	Unweighted Conformal	89.8% [87.4%, 92.4%]	0.054	10.4%	0.045
	SpotGuard (Ours)	89.8% [87.4%, 92.4%]	0.058	11.0%	0.042
Cross-Plat. (10x $\rightarrow$ Slide)	Bayesian Joint CI	1.4% [0.4%, 2.6%]	0.018	–	0.142
	Unweighted Conformal	84.6% [81.4%, 87.8%]	0.062	15.6%	0.061
	SpotGuard (Ours)	89.4% [86.8%, 92.1%]	0.071	18.4%	0.048

the selective gatekeeper allows us to mask 18.4% of spots with 91.3% precision and 88.5% recall against spots deemed out-of-coverage, resulting in a 66.2% error reduction in mean compositional Total Variation error on retained tissue (from 0.142 to 0.048). The Bayesian Joint Confidence Interval reduces to 1.0%–2.0%, as an unadjusted Cartesian product of marginals makes the Bayesian Joint Confidence Interval a hypercube, failing to account for the simplex sum constraint and ignoring compositional covariances, which leads to geometric over-exclusion of the true fractions.

4. Discussion, Limitations, & Access

Clinical Utility & Access. A single scalar uncertainty radius can provide clinically useful gatekeeping information for pathologists on intractable high-dimensional posterior distributions. Intractable covariates motivate tools that reduce them to scalars. **Web App:** <https://the-mathking.github.io/ML4HDemo/> • **Video:** <https://youtu.be/u3NUaWljldgo> • **Code:** <https://github.com/The-MathKing/ML4HDemo>

Limitations & Scope. SpotGuard uses reference single-cell pools to define this uncertainty region, which may not contain all possible rare tumor states. While density-ratio weighting addresses covariate shift in spot gene expression ($P(X)$), extreme biological label shifts ($P(Y|X)$) require ongoing validation against physical multiplex immunofluorescence co-stains.

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